Neutron Detector and Generator Characterization for the BABY Experiment

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LIBRA and BABY Experiments

The LIBRA (Liquid Immersion Blanket: Robust Accountancy) project is a laboratory-scale demonstration of a Liquid Immersion Blanket that breeds tritium using a crucible of molten salt and a DT neutron generator.

Before full-scale LIBRA operations commence, BABY (Build a Better Yield) blanket experiments operated at 1 L salt volumes using molten ClLiF salt and a Thermo-Fisher A325 DT neutron generator.



Diamond Detectors

Diamond detectors are semiconductor devices that generate signals when charged particles create electron-hole pairs in the crystal. Diamonds are well-suited for neutron detection in fusion experiments because they are:

- Radiation-hard due to their crystal structure
- Low-noise due to a wide bandgap, enabling operation at high temperatures
- Capable of fast response due to high electron/hole mobility

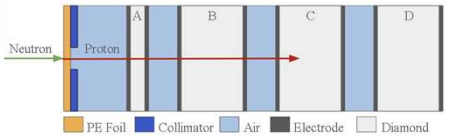
Diamond Proton Recoil Telescope

The BABY experiments used the Cividec A8 Diamond Proton Recoil Telescope, which can detect neutrons by two methods:

- 12C(n,α)9Be nuclear reaction
- Coincidence signals produced by a proton recoiled into the diamond telescope volume by an incident neutron

The coincidence of signals across multiple diamonds filters out background noise and provides a direct reconstruction of the total energy deposited in the detector active volume by the recoiled proton.

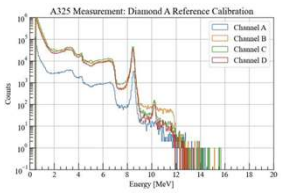


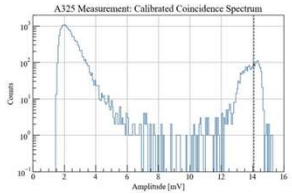


Diamond Detector Characterization

Detector Energy Calibration

- A Ra-226 source was placed next to Diamond A.
- The energy loss due to air and electrode material was estimated using SRIM simulations. This allowed for the determination of the true deposited energy in the detector.
- Once Diamond A was calibrated, it was used as a reference to calibrate Diamonds B, C, and D.
- DT neutron measurements from Diamond A were used to assign true energy values to features in the neutron spectrum.

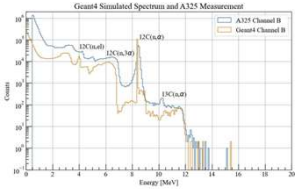




Detector Efficiency Calculation

The total efficiency of the PRT is crucial for translating the detector's count rate into the actual neutron flux, which is key to determining the TBR. To calculate the detector efficiency, Geant4 simulations were used and compared to detector measurements.

The simulated spectrum of neutron energy deposited in the diamond volumes agrees well with the measured spectrum, with all key features represented.





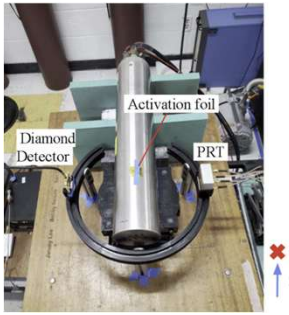
The peak efficiency of the detector was calculated using the number of counts under the 12C(n,α)9Be peak with the continuum removed.

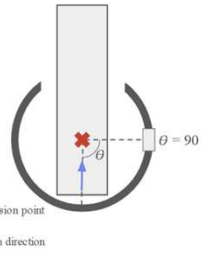
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Neutron Generator Characterization

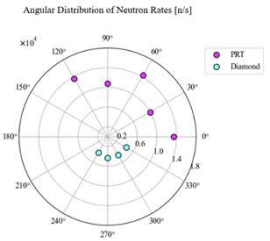
The beam-target nature of the neutron production results in variations in neutron energy and flux relative to the direction of the beamline. An understanding of the neutron environment is crucial for assessing the results of tritium breeding in the molten salt.

Angular Scan





Neutron flux measurements were taken at 9 positions at various angles around the neutron generator using two diamond detectors.



The results of the angular scan indicate angular asymmetry in the neutron rate measured around the generator consistent with expected shielding effects from the generator's internal material. Differences in the rate measured by the two detectors may be due to systematic errors in detector placement.

Kriging was used to interpolate the discrete neutron rate measurements, creating a continuous flux map with associated uncertainty contours. The model's predicted source rate had a 32% relative error compared to the activation foil measurement , which was reduced to a 12% relative error with the addition of a new data point measured in a region of high uncertainty.

